



GOVERNMENT OF KARNATAKA

(Agriculture Produce Market Committee, Channarayapatna)

**Name of the work: Construction of CC Road, RCC Drain
& CAP at Didiga SMY of APMC Channarayapatna.**

Under NABARD RIDF-29 Scheme - RM29052

Head of Account: 2023-24 NABARD RIDF 29

**OFFICE OF THE
Assistant Executive Engineer,
MDP Subdivision No-6
Arsikere**

ESTIMATED COST: 100.00 Lakhs

**Consultant:
GEOPAGE CONSULTANTS**

#104/22, 6th Cross, KHB Colony, 2nd Stage,
Basaveshwaranagar, Bengaluru – 560079

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GOVERNMENT OF KARNATAKA
(Agricultural Marketing Department)

(PARAGRAPH - 13 C (A) OF THE KARNATAKA PUBLIC WORKS
DEPARTMENTAL CODE)

**Name of work: - Construction of CC Road, RCC Drain & CAP at Didiga SMY
of APMC Channarayapatna Under NABARD RIDF-29 Scheme**

FACE SHEET

DIVISION	MDP, MYSORE
SANCTIONED ESTIMATE NO.	NO. DATE
FUND HEAD	NABARD RIDF-29
MAJOR HEAD	
SERVICE HEAD	
DEPARTMENTAL HEAD	
ESTIMATE FRAMED	OFFICE OF THE ASSISTANT EXECUTIVE ENGINEER, M.D.P NO. 6 SUB DIVISION, ARSIKERE
ADMINISTRATIVELY APPROVED UNDERGO No.	Administrative latter No: DAM/SE/MDP/ /2022-23 Date:
TECHNICALLY SANCTIONED	NO. DATE
ESTIMATE PREPARED BY	Geopage Consultants Basaveshwaranagar, Bengaluru – 560079
AND CHECKED BY (CALL OR AUTHORITY)	M. Manjunathaiah (i/c) Assistant Executive Engineer

<u>CHECK LIST</u>		
1	Name of Sandy	Didaga
2	Name of A.P.M.C	Channarayapatna
3	Taluk / District	Channarayapatna / Hassan
4	Extent of land	12.25 acres
5	Govt. Land / Private Land	Govt. Land
6	Details Of Transaction	Vegetables, Food Grains and other daily Consumbale materials
7	Whether adjacent to the roadside or not, if not the distance	Adjacent to the Nuggehalli-Tiptur Road
8	Day of Sandy	Tuesday
9	No. of Population generally gather on sandy day	3960 people
10	No. of Benefitted Villages generally gather on sandy day	154 Villages
11	Where drinking water facilities available	Yes

Assistant Engineer

Asst Executive Engineer
No.6 Subdivision, MDP
Arsikere

INSPECTION OF SITE CERTIFICATE

(As per Para 133 of K.P.W.D) and (As required vide G.O No. PWD SL CMS 96 Dt. 27-06-1991)

Certified that Assistant Executive Engineer, MDP, Sub Division, Arsikere have personally visited the spot on **09-09-2023** and verified the estimate using sanctioned schedule of rates, and providing for the most economical and safe way of executing the work.

This is to certify that the estimate for "**Construction of CC Road, RCC Drain & CAP at Didiga SMY of APMC Channarayapatna**" has been prepared in accordance with procedure laid down in the departmental code using the available rates from Common SR Volume-I, Volume-II, Volume-III issued by KPWD for the year 2023-2024 and with additional cost of 18% GST is considered.

Certified that sufficient care has been taken while preparing the estimates in respect of the above mentioned work to see that necessary items which are technically required for the work in sufficient quantity as per site condition are included while preparing the estimate, and specification of items included are completed and correct.

DUPLICATION OF WORK CERTIFICATE

The said work is not taken-up under any earlier schemes. Certified that the work proposed in this estimate has not been taken up under any other head/ programme by any agency. The said work has not been included/ proposed/ overlapped in any other proposed programme. This work is taken-up under the Head of Account NABRD RIDF 29 for the year 2023-24, at an estimated cost of Rs. **100.00** Lakhs. This work has been approved in the State Action Plan 2023-24 dt:..... at Sl. No.

Section Officer,

Assistant Executive Engineer,
MDP. Sub-Division,
Arsikere

Executive Engineer,
MDP. Division,
Mysore

IMPLEMENTATION SCHEDULE

Name of the work: Construction of CC Road, RCC Drain & CAP at Didiga SMY of APMC Channarayapatna Under NABARD RIDF-29

Item of work	Nov-Dec (2023)	Jan-Feb (2024)	Mar-Apr (2024)	Jun-Jul (2024)
Preparation of Estimate,NABARD approval & Tender Process				
Formation, Utility lines,EW,PCC				
Constuction of Drain, Deck Slab, Constuction of Road & Paver Blocks				
Covered Auction Platform				
Cash Flow in Lakhs	1.00	19.80	49.50	29.70

Section Officer,

Assistant Executive Engineer,
MDP. Sub-Division,
Arsikere

Brief Project Report

The development of agriculture and horticulture has become a major social aspiration in many developing countries, more than ever before. This has manifested in various kinds of advances on agricultural front. Marketing of agriculture and horticulture produce is as important as production itself as it plays a very important role not only in stimulating production and consumption but also in increasing the pace of economic growth. It is however, unfortunate that the marketing sector in the field of agriculture and horticulture has not progressed to the desired extent, in developing countries like India.

Under these circumstances, the efforts put in by the various government agencies like APMC's to promote the regulated agricultural markets deserve to be commended. There has been demand in the past to regulate the working of regulated markets throughout the country. But now the things have changed and regulation now is being replaced by development.

The existing structure of marketing of agriculture and horticulture produces does not address the problems of farmers, As It has low marketing efficiency, high post harvest losses and does not foster competitiveness. A modern innovative system which reduce vested interest of a large intermediary chain, creates competition, assures quality and modernizes operation with IT applications. In handling of agriculture and horticulture produce it is very necessary to raise the income of farmers. To achieve these objectives, APMC of Channarayapatna has decided to make improvements to Main Market Yard.

The following are the major notified commodities Marketing in the Channarayapatna A.P.M.C. yard

1. Coconut
2. Copra
3. Cereals
4. Pulses
5. Ginger
6. Jaggary
7. Maize
8. Horticultural produce
9. Vegetables
10. Cattles

The Agricultural produce market committee Channarayapatna is one of the most important market committees in the state of Karnataka because of its strategic location. It is located besides NH 48 (Bangaluru-Mangaluru highway) and is about 10 kms away from Shravabelagola (International Tourism Spot) and 35 Kms from Hassan. The town enjoys a good Road transport facilities. These factors add to the potentiality of this market. It is both a primary and a secondary market

The Channarayapatna Main Market Yard is having an area of 25-26 Acre of land which is Not yet developed.

The Didaga Sub Market Yard is having an area of 11-10 Acre of land which is Not yet developed.

The total arrivals of Notified Agricultural Commodities viz., Coconuts and Tender coconuts in nos is as follows

Sl.No	Year	Commodities	No.s	value
1	2020-21	Coconut	14704712	173021944
		Tender coconuts	5607375	56113750
2	2021-22	Coconut	6077862	67897560
		Tender coconuts	3151500	34074000
3	2022-23	Coconut	4586226	48022260
		Tender coconuts	392980	4227300

Other notified commodities in quintals. Is as follows

Sl. No	Year	Quintals	value
1	2020-21	225842	1042299919
2	2021-22	29141	140151898
3	2022-23	19624	76669547

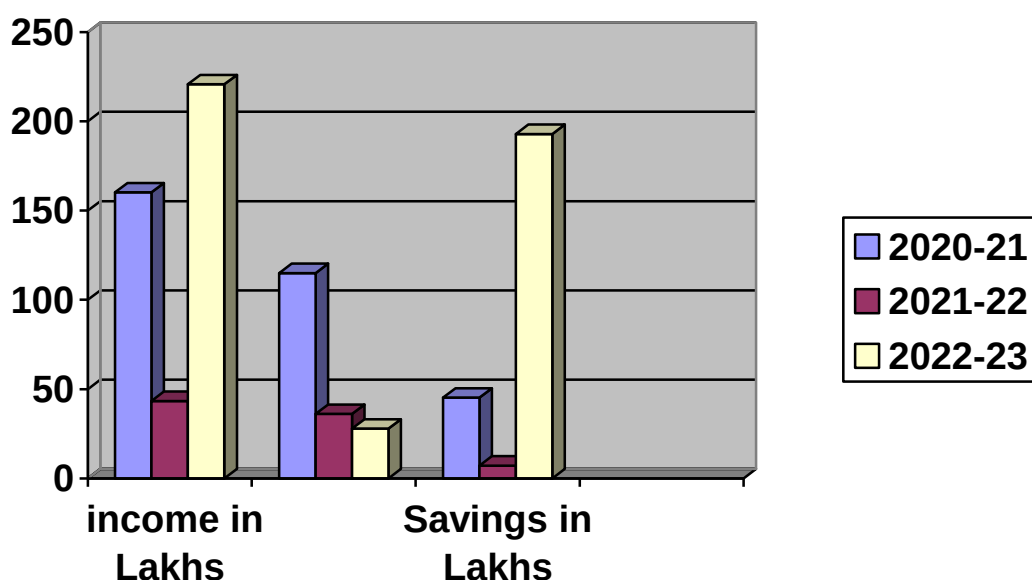
This fact indicates that there is an urgent need to improve and develop Channarayapatna Main Market Yard immediately with modern amenities.

APMC provides service to users of markets viz., farmers, commission agents, traders, importers, exporters and other market functionaries and consumers. It retains the role of coordination and acts as a guide to all the users of the market for the efficient running of the agricultural marketing system.

The Business of food grains and vegetables takes place throughout the year. The Business activities are showing substantial growth every year. This can be seen from the arrival and sales charts. The APMC is experiencing a successive increased income year by year by means of market fees and license fees. The income for the year 2022-23 is Rs. 220.67 lakhs.

INCOME AND EXPENDITURE STATEMENT OF A.P.M.C CHANNARAYAPATNA

Year	Market Income ` In Lakhs	Expenditure ` In Lakhs	Savings ` in Lakhs
2020-21	160.16	114.94	45.22
2021-22	43.34	36.16	7.17
2022-23	220.67	27.84	192.82



**The Proposed work at Construction of CC Road, RCC Drain & CAP at
Didiga SMY of APMC Channarayapatna Under NABARD RIDF-29
Scheme**

sl. No.	Description	Amount (In Rs)
1	Construction of C.C Road along with Drain and Deck Slabs at Didaga Sub-Market Yard of APMC Channarayapatna. .	51,84,830.00
2	Construction of Covered Auction Plat Form at Didaga Sub Market Yard of APMC Channarayapatna.	32,07,155.00
	TOTAL	83,91,985.00
3	GST @ 18%	15,10,557.00
4	DPR Charges @0.45%	37,764.00
5	Third Party Charges @0.5%	41,960.00
6	Miscellaneous and Rounding off	17,734.00
	GRAND TOTAL	1,00,00,000.00

Technical and Administrative approval will be sought from the engineering cell of the Market Development Project and Director of the Agricultural Marketing Department.

The Total cost of the Project with the above proposals is Rs, 100.00 Lakhs and the Marketing Committee has a weak financial available with it and hence it is submitting this project for Assistance from NABARD RIDF-29 Scheme to facilitate Farmers of this Town. The rates for the name of work “Construction of CC Road, RCC Drain & CAP at Didiga SMY of APMC Channarayapatna Under NABARD RIDF-29 Scheme” are taken from Common SR Volume-I, Volume-II, Volume-III issued by KPWD and Volume-V Issued by BWSSB for the year 2023-2024 and area weightage are added accordingly in the preparation of the estimate. With the creation of the proposed infrastructure facilities in the proposed extension market yard and Main market yard it is expected that the Market yard will start functioning and gradually the infrastructure facilities may be improved as the trade increases. This will in turn increase Agricultural Production gradually and therefore all other related activities like trade, export and import of agriculture produce will also increase. Hence it can be safely anticipated that the income will increase @ 20% per annum for the next ten years.

Assistant Executive Engineer
No.6 Sub-Division, MDP
Arsikere

Construction of CC.Road, RCC Drain & CAP at Didiga SMY of APMC
Channarayapatna
ESTIMATED COST-100.00 LAKHS



1st Main Road



CAPF LOCATION

ABSTRACT

**Name of Work :- Construction of CC Road, RCC Drain & CAP at Didiga SMY
of APMC Channarayapatna Under NABARD RIDF-29 Scheme**

sl. No.	Description	Amount (In Rs)
1	Construction of C.C Road along with Drain and Deck Slabs at Didaga Sub-Market Yard of APMC Channarayapatna. .	51,84,830.00
2	Construction of Covered Auction Plat Form at Didaga Sub Market Yard of APMC Channarayapatna.	32,07,155.00
	TOTAL	83,91,985.00
3	GST @ 18%	15,10,557.00
4	DPR Charges @0.45%	37,764.00
5	Third Party Charges @0.5%	41,960.00
6	Miscellaneous and Rounding off	17,734.00
	GRAND TOTAL	1,00,00,000.00

Section Engineer

Assistant Executive Engineer
No. 6 Sub-Division, MDP,
Arsikere.

Executive Engineer
MDP,Mysore Division
Mysore

SUB ESTIMATE
FOR CC PAVEMENT, CC DRAIN
AND DECK SLAB

Detailed Estimate

Name of work:-Construction of CC Road, RCC Drain & CAP at Didiga SMY of APMC Channarayapatna Under NABARD RIDF-29 Scheme

Sl No	Item of work	No	L	B	D	Qty	Rate	Amount
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- 1 Earth work in surface excavation by mechanical means for lowering & leveling the ground for all works other than foundation in all kinds of soils & upto depth not exceeding 300mm as per drawing and technical specifications, including setting out, shoring, strutting, barricading, caution lights, removal of stumps and other deleterious matter including dressing of excavated surfaces, disposing off or levelling the excavated earth or sorting & stacking the selected earth for reuse in a radius of 50 m and lift upto 1.5 m including cost of labour, tools, usage of machinery & other appurtenances required to complete the work.. **In all kinds of soils upto 300 mm depth**

(KPWD-SR 2023-24, Vol-1, Page No. 8, Item No.1.12)

Rate Analysis

Finised Rate	95.00
Area waitage 3%	2.85
Net rate	97.85

To R.C.C Drain

1st Main Road	1	2	109.00	1.05	$\frac{0.25+0.10}{2}$	40.06
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To C.C Road

Inbetween Drain

1st Main Road	1	109.00	13.20	$\frac{0.35+0.10}{2}$	323.73
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Total	363.79	97.85	35597.00
	Cum		

- 2 Earth work excavation for Foundation by mechanical means for all works & depth up to 3 m, as per drawing and technical specifications, including setting out, shoring, strutting, barricading, caution lights, including dressing of excavated surfaces, disposing off or levelling the excavated earth or sorting & stacking the selected earth for reuse in a radius of 50 m and lift up to 1.5 m including cost of labour, tools, usage of machinery & other appurtenances required to complete the work **In all kinds of soils Depth up to 3 m**

(KPWD-SR 2023-24; Vol 1; Page No 8; Item No. 1.14.1)

Rate Analysis

Finised Rate	102.00
Area waitage 3%	3.06
Net rate	105.06

To R.C.C Drain

1st Main Road	2	109.00	1.05	0.70	160.23
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To C.C Road

1st Main Road	1	109.00	11.20	0.45	549.36
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Cement Concrete Pipe

1st Main Road	4	11.70	0.70	0.80	26.20
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To INSPECTION CHAMBER

1st Main Road	8	0.91	0.91	1.00	6.62
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For Paver Block

1st Main Road	2	109.00	1.00	0.24	52.32
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Total	794.73	105.06	83494.00
	Cum		

Sl No	Item of work	No	L	B	D	Qty	Rate	Amount
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- 2 Providing and laying of a **reinforced cement concrete pipe duct, 300 mm dia**, across the road (new construction), extending from drain to drain in cuts and toe of slope to toe of slope in fills, constructing head walls at both ends, providing a minimum fill of granular material over top and sides of RCC pipe as per IRC:98-1997, bedded on a 0.3 m thick layer of granular material free of rock pieces, outer to outer distance of pipe at least half dia of pipe subject to minimum 450 mm in case of double and triple row ducts, joints to be made leak proof, invert level of duct to be above higher than ground level to prevent entry of water and dirt, all as per IRC: 98 - 1997 and approved drawings. **Single Row for one utility service.**

(KPWD-SR 2023-24; Vol 3; Page No 71; Item No.8.29 (i))

Rate Analysis

Finised Rate	2212.00
Area waitage 3%	66.36
Net rate	2278.36

Cement Concrete Pipe

1st Main Road	4	11.70	-	-	46.80			
					Total	46.80	2278.36	106627.00
						mtr		

- 3 Providing and laying in position cement concrete for levelling course for all works in foundation. The granite/trap/basalt crushed graded coarse aggregates and fine aggregates as per relevant IS Codes machine mixed, laid in layers not exceeding 150 mm thickness, well compacted using plate vibrators, including all lead & lifts, cost of all materials of quality, labour, Usage charges of machineries, curing, and all the other appurtenances required to complete the work as per technical specifications. **Mix 1:3:6 (M10) Using 40 mm and down size graded crushed coarse aggregates.**

(KPWD-SR 2023-24; Vol 1; Page No 15; Item No. 2.1.3)

Rate Analysis

Finised Rate	5989.00
Area waitage 3%	179.67
Net rate	6168.67

To R.C.C Drain

1st Main Road	2	109.00	1.05	0.10	22.89			
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To INSPECTION CHAMBER

1st Main Road	8	0.91	0.91	0.15	0.99			
					Total	23.88	6168.67	147308.00
						Cum		

- 4 Providing **Brick work with common burnt** clay Non Modular bricks of class designation 3.5 in foundation and plinth in Cement mortar 1:6 (1 cement : 6 coarse sand) including cost of all materials, labour, scaffolding and usage charges of machinery & other incidental charges complete as per the direction of engineer incharge of work...

(KPWD-SR 2023-24; Vol 2; Page No 13; Item No.6.2)

Rate Analysis

Finised Rate	7978.00
Area waitage 3%	239.34
Net rate	8217.34

To INSPECTION CHAMBER

1st Main Road	8	2.72	0.23	1.05	5.25			
					Total	5.25	8217.34	43141.00
						Cum		

Sl No	Item of work	No	L	B	D	Qty	Rate	Amount
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- 5 Providing and laying in position cement concrete for all Super structures of building , Road works, Water works, Irrigation works & super structure works of bridges upto 3.50 m height. The granite/ trap/basalt crushed graded coarse aggregates and fine aggregates as per relevant IS Codes machine mixed with super plasticisers laid in layers, well compacted using needle vibrators. The cost includes all lead & lifts, cost of all materials, quality confirming to the requirements of relevant IS codes , labour, Usage charges of machinery, curing and all other appurtenances required to complete the work as per technical specifications. **The M25 Design Mix Using 20 mm nominal size graded crushed coarse aggregates.**

(KPWD-SR 2023-24; Vol 1; Page No 16; Item No.2.5.2)

<u>Rate Analysis</u>	
Finised Rate	6843.00
Area waitage 3%	205.29
Net rate	7048.29

To INSPECTION CHAMBER COVER

1st Main Road	8	0.91	0.91	0.20	1.32			
						Total	1.32	7048.29
							Cum	9304.00

- 6 Providing and laying in position cement concrete for all Foundation works. The granite/trap/basalt crushed graded coarse aggregates and fine aggregates as per relevant IS Codes machine mixed with super plasticizers laid in finished layers, well compacted using needle vibrators, including all lead & lifts, cost of all materials, quality confirming to the requirements of relevant IS codes, labour, Usage charges of machinery, curing and all the other appurtenances required to complete the work as per technical specifications .**M20 Design Mix Using 20 mm and down size graded crushed coarse aggregates.**

(KPWD-SR 2023-24; Vol 1; Page No 15; Item No. 2.2.1)

<u>Rate Analysis</u>	
Finised Rate	6287.00
Centering @ 35%	2200.45
Area waitage 3%	188.61
Net rate	8676.06

To R.C.C Drain

1st Main Road								
Wall	1	4	109.00	0.15	0.60	39.24		
						Total	39.24	8676.06
							Cum	340449.00

- 7 Providing and laying in position cement concrete for all Foundation works. The granite/trap/basalt crushed graded coarse aggregates and fine aggregates as per relevant IS Codes machine mixed with super plasticizers laid in finished layers, well compacted using needle vibrators, including all lead & lifts, cost of all materials, quality confirming to the requirements of relevant IS codes, labour, Usage charges of machinery, curing and all the other appurtenances required to complete the work as per technical specifications .**M20 Design Mix Using 20 mm and down size graded crushed coarse aggregates.**

(KPWD-SR 2023-24; Vol 1; Page No 15; Item No. 2.2.1)

<u>Rate Analysis</u>	
Finised Rate	6287.00
Centering @ 5%	314.35
Area waitage 3%	188.61
Net rate	6789.96

To R.C.C Drain

1st Main Road								
Bed	2	109.00	0.90	0.15	29.43			

Sl No	Item of work	No	L	B	D	Qty	Rate	Amount
Cover Deck								
	1st Main Road	2	109.00	0.90	0.15	29.43		
Deck Slab								
	Road Crossing	2	13.20	0.90	0.20	4.75		
Total						63.61	6789.96	431909.00
						Cum		

- 8 Supplying, fitting and placing TMT FE 550 / 550D Steel Reinforcement including cost of all materials, machinery, labour, cleaning, straightening, cutting, bending, hooking, laping/welding joints, tying with binding wire / soft annealed steel wire and other ancillary operations complete as per drawing and technical specification

(KPWD-SR 2023-24; Vol 1; Page No 18; Item No. 2.11 h)

Rate Analysis

Finised Rate	79011.00
Area waitage 3%	2370.33
Net rate	81381.33

To INSPECTION CHAMBER

TOP SLAB				22.5				
				Rmtr				
	at 22.5 Kg/rmtr					506.25		
RCC Drain				218.00				
				Rmtr				
	at 21.00 Kg/rmtr					4578.00		
Deck Slab				13.20				
				Rmtr				
	at 22.5 Kg/rmtr					297.00		
						5381.25		
Total						5.38	81381.33	437933.00
						T		

- 11 Providing **weep holes** in Brick masonry/Plain/ Reinforced concrete abutment, wing wall/ return wall with 100 mm dia AC pipe, extending through the full width of the structure with slope of 1V :20H towards drawing face. Complete as per drawing and Technical Specifications

KPWD-SR 2023-24; Vol 3; Page No 117; Item No. 12.1

Rate Analysis

Finised Rate	138.00
Area waitage 3%	4.14
Net rate	142.14

To R.C.C Drain

1st Main Road	1	24	0.15		3.60			
Total						3.60	142.14	512.00
						M		

Sl No	Item of work	No	L	B	D	Qty	Rate	Amount
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Cement Concrete Road

12 Compacting Original Ground, Compacting original ground supporting embankment

Loosening, leveling and Compacting original ground supporting embankment to facilitate placement of first layer of embankment, scarified to a depth of 150 mm, mixed with water at OMC and then compacted by rolling so as to achieve minimum dry density as given in Table 300-2 for embankment construction.

(KPWD-SR 2023-24; Vol 3; Page No 17; Item No. 3.7)

Rate Analysis

Finised Rate	105.00
Area waitage 3%	3.15
Net rate	108.15

To C.C Road

1st Main Road	1	109.00	11.20	0.10	122.08			
				Total	122.08	108.15	13202.00	
					Cum			

13 Construction of Granular Sub-Base of required grading as per design mixing in a mechanical mix plant at OMC, carriage of mixed Material to work site, spreading in uniform layers with motor grader on prepared surface and compacting with vibratory power roller to achieve the desired density, complete as per clause 401

(v)For Grading -V Material

(KPWD-SR 2023-24; Vol 3; Page No 23; Item No. 4.1 A(V))

Rate Analysis

Finised Rate	2487.00
Area waitage 3%	74.61
Net rate	2561.61

To C.C Road

1st Main Road	1	109.00	11.20	0.15	183.12			
				Total	183.12	2561.61	469082.00	
					Cum			

14 Dry Lean Cement Concrete Sub- base

Construction of dry lean cement concrete Sub- base over a prepared sub-grade with coarse and fine aggregate conforming to IS: 383, the size of coarse aggregate not exceeding 25 mm, aggregate cement ratio not to exceed 14:1, aggregate gradation after blending to be as per table 600-1, cement content not to be less than 140 kg/ m3, optimum moisture content to be determined during trial length construction, concrete strength not to be less than 7 Mpa at 7 days, mixed in a batching plant, transported to site, laid with a paver with electronic sensor, compacting with 8-10 t vibratory roller, finishing and curing.

(KPWD-SR 2023-24; Vol 3; Page No. 49; Item No. 6.1)

Rate Analysis

Finised Rate	4920.00
Area waitage 3%	147.60
Net rate	5067.60

To C.C Road

1st Main Road	1	109.00	11.20	0.10	122.08			
For Paver Block								
1st Main Road	2	109.00	1.00	0.10	21.80			
				Total	143.88	5067.60	729126.00	
					Cum			

Sl No	Item of work	No	L	B	D	Qty	Rate	Amount
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- 15 **Plain Cement Pavement Quality M30 grade Concrete (RMC)** with OPC & GGBS laid without Paver. Construction of un-reinforced plain cement concrete pavement using RMC M-30 grade with OPC @ 270 kg/m³ and GGBS @ 90 kg/m³ (75:25 proportion) as per approved mix design procedure and thickness as per design, over a prepared sub base. The superplasticiser confirming to IS 9103-1999 Reaffirmed-2008, Coarse aggregates and Fine aggregate confirming to IS:383-2016, transported to site, laid in approved fixed side form work (steel channel, laying and fixing of 125 micron thick polythene film, wedges, steel plates including levelling the form work). Spreading the concrete with shovels, rammers and compacted using needle, suitable plate vibrator and finished in a continuous operation including provision of separation membrane and Hessian cloth finishing to lines and grades complete including cost of all materials, labour, all lead & lift, loading charges as per specification & direction of Engineer - incharge of the work. (Groove cutting shall be paid separately)

(KPWD-SR 2023-24; Vol 3; Page No 51; Item No. 6.4)

<u>Rate Analysis</u>	
Finised Rate	6774.00
Area waitage 3%	203.22
Net rate	6977.22

To C.C Road

1st Main Road	1	109.00	11.20	0.20	244.16			
				Total	244.16	6977.22	1703558.00	
					Cum			

- 16 Providing and fixing in position **25mm dia dowel bars** precoated with anticorrosive epoxy paint of required Dia. 600 mm. Long and at 300 mm. C/C and wherever directed including handling, straightening, necessary cutting supported by TMT FE 500, chairs with proper alignment by using properly designed assembly of Bulkheads lubricating half length with bituminous paint as directed etc. complete.

(KPWD-SR 2023-24; Vol 3; Page No 163; Item No. 17-C 2)

<u>Rate Analysis</u>	
Finised Rate	380.00
Area waitage 3%	11.40
Net rate	391.40

To C.C Road

1st Main Road	411				411.00			
				Total	411.00	391.40	160865.00	
					No			

- 17 Providing and fixing in position, **tie bars** precoated with anticorrosive epoxy paint of 12 mm dia. 700 mm. long and at 300 mm. C/C and wherever directed including handling, straightening wrapping with paper of approved quality for half length, necessary cutting, handling, straightening, supported by assembly of TMT FE 500, chairs with proper alignment etc. complete.

(KPWD-SR 2023-24; Vol 3; Page No 163; Item No. 17-C 3)

<u>Rate Analysis</u>	
Finised Rate	270.00
Area waitage 3%	8.10
Net rate	278.10

To C.C Road

1st Main Road	546				546.00			
				Total	546.00	278.10	151843.00	
					No			

Sl No	Item of work	No	L	B	D	Qty	Rate	Amount
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- 18 Cutting transverse contraction joints 3 to 4 mm wide and depth 60mm in concrete slab using concrete cutting machine with diamond studded saw within 4-8 hours of casting of bay / slab etc. complete including subsequent widening of the groove 8 to 10 mm. wide at top having depth of 15 mm. as directed by Engineer incharge

(KPWD-SR 2023-24; Vol 3; Page No 163; Item No. 17-C-4)

Rate Analysis

Finised Rate	70.00
Area waitage 3%	2.10
Net rate	72.10

To C.C Road

1st Main Road	244.16	244.16
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Transverse cutting @5m interval

Total	244.16	72.10	17604.00
	M		

- 19 Providing and laying 100mm thick factory made precast M -40 grade Cement Concrete Paver Block as per IRC SP 63:2018 & IS15658 for Commercial Traffic Axle load repetitions more than 10MSA upto 20MSA for Industrial Streets, Bus & Truck parking of approved shape and colour, laid in required pattern and including over 40mm thick compacted bed of coarse sand, filling the joints with fine sand etc. all complete as per the direction of Engineer-in-charge. (WMM/WBM Base to be paid separately as per relevant technical specification)

(KPWD-SR 2023-24; Vol 3; Page No 52; Item No. 6.8.3)

Rate Analysis

Finised Rate	1300.00
Area waitage 3%	39.00
Net rate	1339.00

For Paver Block

1st Main Road	2	109	1.0	218.00		
Total				218.00	1339.00	291902.00
				Sqm		

- 20 Providing, supplying, fabricating and fixing in position Name board of size 1.60 M x 1.00 M made of M.S.angles of size 50 x 50 x 6mm for outer frame and for cross bracings and 35 x 35 x 5 mm for 1 No. horizontal and vertical supports as per the drawing and details and including providing and fixing with plain aluminium sheet on top of M.S.frame work including using reflector sheet for backing as well as lettering of appropriate size as per the required text to be lettered including fixing the board to wall, column as directed at site etc., complete as per the special speifications and directions of the Engineer-In-Charge of the work and inclusive of cost

Rate Analysis

Finised Rate	11374.00 /Each (SE approved rate)
Net rate	11374.00

Name Board	1	-	-	-	1.00	
Total					1.00	11374.00
					Nos	11374.00

Total 5184830.00

Section Engineer

Assistant Executive Engineer
No. 6 Sub-Division, MDP,
Arsikere.

Executive Engineer
MDP,Mysore Division
Mysore

DESIGN OF CEMENT CONCRETE PAVEMENT

Name of Work: Construction of CC.Road, RCC Drain & CAP at Didiga SMY of APMC Channarayapatna

Ref: Design is based on 'IRC 58: 2015-Design of Plain Jointed Rigid Pavements for Highways'

1. Data:

A Cement Concrete Pavement Road is to be designed for Roads in APMC **Didiga SMY of APMC Channarayapatna** where traffic volume is **496** Commercial vehicles per day Consisting of Agricultural tractor/trailers, light goods vehicles, heavy trucks etc. In addition, heavy trucks park their vehicles along the road before they move to Go downs to carry out loading and unloading activities.

2. Design:

Wheel Load, the Wheel load appropriate for the traffic conditions is 100 KN. Sub-grade/Formation Soil Shall have CBR Value of **7.8%** and it Shall have thickness to suit the formation and Profile of the Road. The existing Subbase is to be prepared and compacted to achieve the required degree of Compaction. Over Sub-graded GSB with CBR Value of 25% (Minimum) is Proposed. An additional layer 100mm DLC (Dry Lean Concrete) as foundation layer is also proposed to cater the static load of heavy trucks.

Construction of CC.Road, RCC Drain & CAP at Didiga SMY of APMC Channarayapatna			
IRC:58-2015 Guidelines for Design of Plain Jointed Rigid Pavements for Highways			
Design of Slab Thickness for Pavement			
Type of pavement considered		Pavement Structural Details	
Carriageway	2/4-lane	Modulus of subgrade reaction of subgrade, MPa/m	49
CBR (%)	7.8	Thickness of Granular Subbase, mm	150
Shoulders :- Tied concrete shoulders ? (yes/no)	No	Thickness of Dry Lean Concrete subbase, mm	100
Transverse joint spacing (m)	10	Effective modulus of subgrade reaction of foundation, MPa/m	218
Lane width (m)	3.5	Unit weight of Concrete, kN/m ³	24
Transverse Joints have dowel bars? (yes/no)	yes	28-day Flexural strength of cement concrete, MPa	4.5
Design Traffic Estimation		Max. day-time Temperature Differential in slab, °C (for bottom-up cracking)	19
Design Period (years)	20	Night-time Temperature Differential in slab, °C (for top-down cracking) = day-time diff/2 + 5	14.5
Total Two-way Commercial Traffic (cvpd) in the year of completion of construction	496	Trial Thickness of Concrete Slab, m	0.20
Av. Annual rate of growth of commercial traffic (expressed as decimal)	0.05	Load Transfer Efficiency Factor for TDC analysis, Beta = 0.66 for dowel Joints, 0.90 for joints without dowels	0.66
Cumulative No of Commercial vehicles during design period (two-way), A	5986260	Elastic Modulus of Concrete, Ec (MPa)	30000
Average No of axles per commercial vehicle, B	2.65	Poisson's Ratio of Concrete, Mu	0.15
Cumulative No of Commercial Axles during design period (two-way), C = A*B	15863590	Radius of relative stiffness, m	0.55350
Proportion of traffic in predominant direction (For 2-lane 2-way highways use a value of), D	0.25	Design Axle Load Repetitions for Fatigue Analysis	
Number of axles in predominant direction	3965897	For Bottom-up Cracking Analysis	
Lateral Placement factor , E	0.250	Front single (steering) Axles = H * K1	89233
Design traffic after adjusting for lateral placement axles	991474	Rear single Axles = H * K2	29744
Night time (12 hr) design axle repetitions (60% traffic during night time)	594884	Tandem Axles = H * K3	49574
Day time (12 hr) design axle repetitions (40% traffic during day time)	396590	Tridem Axles = H * K4	29744
Day time six hr axle load repitions	198295	For Top-Down Cracking Analysis	
Design axle repetitions for BUC analysis (for 6 hour day time traffic), H =	198295	Front single (steering) Axles = J * K1	49078
Proportion of vehicles with spacing between front and the first rear axle less than the spacing of transverse joints, I	0.55	Rear single Axles = J * K2	16359
Design axle repetitions for TDC analysis (for 6-hour night time traffic), J = H*I	109062	Tandem Axles = J * K3	27266
Proportion of Front single (steering) Axles, K1	0.45	Tridem Axles = J * K4	16359
Proportion of Rear single Axles, K2	0.15		
Proportion of tandem Axles, K3	0.25		
Proportion of Tridem Axles, K4 = (1-K1-K2-K3)	0.150		

Fatigue Damage Analysis																		Fatigue Damage Analysis																		
Axle Load Spectrum Data									Fatigue Damage Analysis									Fatigue Damage Analysis																		
									Bottom-up Cracking Fatigue Analysis for Day-time (6 hour) traffic and Positive Temperature Differential									Top-Down Cracking Fatigue Analysis for Night-time (6 hour) traffic and Negative Temperature Differential																		
Rear Single Axle			Rear Tandem Axle			Rear Tridem Axle			Rear Single Axles					Rear Tandem Axles				Rear Single Axles					Rear Tandem Axles					Rear Tridem Axles								
Load Group (kN)	Mid-Point of Load Group (kN)	Frequency (%)	Load Group (kN)	Mid-Point of Load Group (kN)	Frequency (%)	Load Group (kN)	Mid-Point of Load Group (kN)	Frequency (%)	Expected Repetitions (ni)	Flex Stress MPa	Stress Ratio (SR)	Allowable Repetitions (Ni)	Fatigue Damage (ni/Ni)	Expected Repetitions (ni)	Flex Stress MPa	Stress Ratio (SR)	Allowable Repetitions (Ni)	Fatigue Damage (ni/Ni)	Expected Repetitions (ni)	Flex Stress MPa	Stress Ratio (SR)	Allowable Repetitions (Ni)	Fatigue Damage (ni/Ni)	Expected Repetitions (ni)	Flex Stress MPa	Stress Ratio (SR)	Allowable Repetitions (Ni)	Fatigue Damage (ni/Ni)	Expected Repetitions (ni)	Flex Stress MPa	Stress Ratio (SR)	Allowable Repetitions (Ni)	Fatigue Damage (ni/Ni)			
185-195	190	0.00	380 - 400	390	0.00	530-560	545	0.00	0	5.450	1.101	0	0.000	0	4.55	0.920	4.25763	0.000	0	3.528	0.713	1348	0.000	0	3.58	0.724	993	0.000	0	3.437	0.694	2244	0.000			
175-185	180	0.00	360 - 380	370	0.00	500-530	515	0.00	0	5.215	1.053	0	0.000	0	4.36	0.882	12.2876	0.000	0	3.419	0.691	2484	0.000	0	3.47	0.702	1830	0.000	0	3.328	0.672	4134	0.000			
165-175	170	0.00	340 - 360	350	0.00	470-500	485	0.00	0	4.979	1.006	0	0.000	0	4.18	0.843	35.462	0.000	0	3.310	0.669	4577	0.000	0	3.36	0.680	3372	0.000	0	3.219	0.650	7616	0.000			
155-165	160	0.00	320 - 340	330	0.00	440-470	455	0.00	0	4.744	0.958	1	0.000	0	3.99	0.805	102.343	0.000	0	3.201	0.647	8433	0.000	0	3.26	0.658	6213	0.000	0	3.111	0.628	14032	0.000			
145-155	150	0.00	300 - 320	310	0.00	410-440	425	0.00	0	4.508	0.911	5	0.000	0	3.8	0.767	295.364	0.000	0	3.093	0.625	15537	0.000	0	3.15	0.636	11446	0.000	0	3.002	0.606	25853	0.000			
135-145	140	0.00	280 - 300	290	0.00	380-410	395	0.00	0	4.272	0.863	21	0.000	0	3.61	0.729	852.423	0.000	0	2.984	0.603	28625	0.000	0	3.04	0.614	21089	0.000	0	2.893	0.584	47632	0.000			
125-135	130	0.00	260 - 280	270	0.00	350-380	365	0.00	0	4.037	0.816	77	0.000	0	3.42	0.691	2460.1	0.000	0	2.875	0.581	52739	0.000	0	2.93	0.592	38854	0.000	0	2.784	0.562	87758	0.000			
115-125	120	0.00	240 - 260	250	0.43	320-350	335	0.00	0	3.801	0.768	290	0.000	211	3.23	0.653	7099.86	0.030	0	2.766	0.559	97167	0.000	116	2.82	0.570	71586	0.002	0	2.676	0.541	163920	0.000			
105-115	110	0.92	220 - 240	230	0.00	290-320	305	0.00	273	3.566	0.720	1088	0.251	0	3.04	0.615	20490.3	0.000	150	2.657	0.537	183479	0.001	0	2.71	0.548	132287	0.000	0	2.567	0.519	344685	0.000			
95-105	100	0.92	200 - 220	210	1.28	260-290	275	0.00	273	3.330	0.673	4088	0.067	633	2.85	0.577	59135	0.011	150	2.549	0.515	397332	0.000	348	2.6	0.526	263922	0.001	0	2.458	0.497	903499	0.000			
85-95	90	1.38	180 - 200	190	3.40	230-260	245	4.65	409	3.095	0.625	15357	0.027	1688	2.67	0.539	173937	0.010	225	2.440	0.493	1095124	0.000	928	2.49	0.504	634310	0.001	761	2.349	0.475	3564736	0.000			
< 85	80	96.79	< 180	170	94.89	< 230	215	95.35	28789	2.859	0.578	57689	0.499	47042	2.48	0.500	745100	0.063	15834	2.331	0.471	4799545	0.003	25873	2.39	0.482	2110404	0.012	15598	2.24	0.453	39770085	0.000			
		100			100			100	29744	Fat Dam from Sing. Axles = 0.843				49574	Fat Dam from Tand			0.113	16359	Fat Dam from Sing. Axles				0.005	27266	Fat Dam from Tand Axles				0.017	16359	Fat Dam from Tridem Axles				0.001
Front Single Axles and Rear Tridem axles not considered for bottom-up analysis									Total Bottom-up Fatigue Damage due to single and tandem axle loads =					0.843	+	0.113	=	0.960	Total Top-Down Fatigue Damage =										0.005	+	0.017	+	0.001	=	0.022	
																		Sum of CFD for BUC & TDC=					0.98													
																		DESIGN IS SAFE SINCE SUM OF CFD FOR BUC AND TDC <= 1																		

Hence, Proposed

Design PQC thickness	200 mm
Design DLC Thickness	100 mm
Design GSB Thickness	150 mm

$$F_b = \frac{(101.6 - b_d)f_{ck}}{95.25} \quad \dots (15)$$

Where,

- F_b = allowable bearing stress, MPa
 b_d = dowel diameter, mm
 f_{ck} = characteristic compressive strength of the concrete, MPa
 (For M 40 concrete, $f_{ck} = 40$ MPa (28 days)
 = 48 MPa (90 days, MORTH))

7.2.5 Since the initial Load Transfer Efficiency (LTE) at the transverse joint is almost 100% and it takes a long time for the LTE to decrease with traffic repetitions, 90-day compressive strength can safely be used for the computation of allowable bearing stress.

7.2.6 For heavy traffic, greater than 450 CVPD, dowels are to be provided at the contraction joints since aggregate inter-lock cannot be relied upon to effect load transfer across the joint to prevent faulting due to the repeated loading of heavy axles. Joint widths of 5 mm and 20 mm may be taken for stress computation in dowel bar at contraction and expansion joint respectively. Recommended diameter and length of dowel bars are given in **Table 5**.

Table 5 Recommended Dimensions of Dowel Bars

Slab Thickness mm	Dowel Bar Details		
	Diameter, mm	Length, mm	Spacing, mm
200	25	360	300
230	30	400	300
250	32	450	300
280	36	450	300
300	38	500	300
350	38	500	300

Note: The values given are for general guidance. Field performance under heavy loading prevalent in India will be the most appropriate guide. Dowel bars are not satisfactory for slabs of small thickness and shall not be provided for slab of less than 200 mm thickness.

7.2.7 Dowel group action : When loads are applied at a joint, a portion of the load is transferred to the other side of the slab through the dowel bars. If the load is near the joint of a pavement slab tied to a concrete shoulder, a part of the load is transferred to the shoulder also. The dowel bar immediately below the wheel load carries maximum amount of load and other dowel bars transfer progressively smaller magnitudes of loads. Repeated loading causes some looseness between the dowel bars and the concrete slab and recent studies indicate that the dowel bars within a distance of one radius of relative stiffness ($1.0 l$) from the point of load application participate in load transfer. Assuming a linear variation of the load carried by different dowel bars within $1.0 l$, the maximum load carried by a dowel bar can be computed. **Appendix VIII** contains an example of design of dowel bar system.

inaccuracy that may occur in the placement during construction. An example of design of tie bar is given in **appendix IX**.

8.2.5 Typical tie bar details for use at central longitudinal joint in double-lane rigid pavements with a lane width of 3.50 m are given in **Table 6**. The same specifications may be used for the tied concrete shoulder also.

Table 6 Details of Tie Bars for Longitudinal Joint of Rigid Pavements

Slab Thickness mm	Tie Bar Details				
	Diameter (d) mm	Max. Spacing, mm		Minimum Length, mm	
		Plain	Deformed	Plain	Deformed
150	8	330	530	440	480
	10	520	830	510	560
200	10	390	620	510	560
	12	560	900	580	640
250	12	450	720	580	640
300	12	370	600	580	640
	16	660	1060	720	800
350	12	320	510	580	640
	16	570	910	720	800

Note: The recommended details are based on the following values of different design parameters: $S_{st} = 125$ MPa for plain bars, 200 MPa for deformed bars; bond stress for plain bars = 1.75 MPa, for deformed bars = 2.46 MPa as per IRC:15. Tie bars deformed/plain shall conform to IS 1786 and IS 432 respectively.

9 REINFORCEMENT IN CONCRETE SLAB TO CONTROL CRACKING

9.1 Jointed reinforced concrete pavement is adopted for longer slabs with transverse joint spacing greater than 5.0. Reinforcement in concrete pavements, is intended to hold the cracked faces tightly together, so as to prevent opening of the cracks and to maintain aggregate interlock required for load transfer. It does not increase the flexural strength of unbroken slab when used in quantities which are considered economical.

9.2 Reinforcement in concrete slabs counteracts the tensile stresses caused by shrinkage and contraction due to temperature or moisture changes. The maximum tension in the steel across the crack equals the force required to overcome friction between the pavement and its foundation, from the crack to the nearest joint or free edge. This force is the greatest in the middle of the slab where the cracks occur first. Reinforcement is designed for this critical location. However, for practical reasons reinforcement is kept uniform throughout the length for short slabs.

9.3 The amount of longitudinal and transverse steel required per m width or length of slab is computed by the following formula:

$$A_s = \frac{L_d f W}{2 S_{st}} \quad \dots (18)$$

in which

A_s = area of steel in mm² required per m width or length of slab

- (v) IS:5500 (Part 2):2004 Vibratory Roller General requirements Part 1: Self Propelled Single Drum
- (vi) IRC:57 “Recommended Practice for sealing of Joins in Concrete Pavements”

6. PREPARATION OF SUBGRADE, SUB-BASE AND LAYING OF SEPARATION MEMBRANE

6.1 General

The embankment/subgrade shall be prepared as under: The existing surface shall be cleared and grubbed. The two passes of roller shall be given after light spray of water. The initial levels will be recorded. The embankment shall be laid as per specifications.

The Pavement Quality Concrete (PQC) shall be laid on two layers of sub-base. Granular sub-base (GSB) acting as a lower sub-base and Dry Lean Concrete (DLC) acting as upper sub-base. GSB, the lower sub-base shall be laid over a subgrade of minimum 500 mm compacted thicknesses. Subgrade shall be of selected earth complying the following requirements:

- (i) No soft spots are present in the subgrade.
- (ii) The subgrade shall have a minimum CBR of 8 per cent.
- (iii) The camber and super-elevation of subgrade shall be same as that of the DLC & PQC.

6.2 Capillary Cut-off

6.2.1 As a result of migration of water by capillarity from the high water table, the soil immediately below the pavement gets more and more wet and this leads to gradual loss in its bearing capacity besides non-uniform support. Several measures, such as, lowering the sub-soil water table by drainage measures, raising of the embankment or provision of a capillary cut-off may be provided for mitigating this deficiency. It should be investigated for arriving at the optimum solution.

6.2.2 The capillary cut-off may be a layer of coarse or fine sand, graded gravel, bituminised material, geo-composite, or an impermeable membrane.

6.2.3 Capillary cut-off/blanket layer, of required thickness may be placed over compacted subgrade layer. Cut-off with bituminised or other materials may be provided in any of the following ways:

(i) Coarse or fine sand or graded gravel

The minimum thickness of these materials shall be 150 mm. Whenever, sand/gravel is used as cut-off layer, the layer shall not be provided at the edges but should be replaced with suitable filter of graded granular material with or without non-woven geo-textile material.

SUB ESTIMATE
FOR CAP

Detailed Estimate of CAP

**Name of Work:- Construction of CC Road, RCC Drain & CAP at Didiga SMY of APMC
Channarayapatna Under NABARD RIDF-29 Scheme
(Each C.A.P.F Size 36.30 x 9.30 Mtr)**

SI No	Item of work	No	L	B	D	Qty	Rate Rs	Amount Rs
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- 1 Earth work in surface excavation for stripping, seating of bund, Road way, by manual means for lowering & leveling the ground for all works other than foundation & depth in all kinds of soil not exceeding 300mm as per drawing and technical specifications, including setting out, shoring, strutting, barricading, caution lights, removal of stumps and other deleterious matter including dressing of excavated surfaces, disposing off or levelling the excavated earth or sorting & stacking the selected earth for reuse in a radius of 50 m and lift upto 1.5 m including cost of labour, tools, usage & other appurtenances required to complete the work. **(KPWD SR:2023-24, Vol-1, Page No. 5, Item No.1.1)**

Levelling Ground	1	40.00	15.00	0.20	120.00			
				Total	120.00		93.73	11248.00
					Cum			

Rate Analysis

Basic Rate	91.00
Area waitage 3%	2.73
Net Rate Rs.	93.73 /Cum

- 2 Earth work excavation by manual means for drains, canals, waste weir, draft, approach channels, key trenches, foundation of bridges and such similar works in all kinds of soils, as per drawing and technical specifications, including setting out, shoring, strutting, barricading, caution lights, removal of stumps and other deleterious matter, excavated surface leveled and sides neatly dressed disposing off the excavated stuff or sorting & stacking the selected stuff for reuse in a radius of 50 m and lift upto 1.5 m including cost of labour, tools & other appurtenances required to complete the work. In all kinds of soils Depth upto 1.5 m **(KPWD SR:2023-24, Vol-1, Page No. 5, Item No.1.4.1)**

Wall Foundation

Long wall	2	36.00	1.00	1.15	82.80
Short wall	2	9.00	1.00	1.15	20.70

Flagging

Long Side	2	38.45	1.00	0.10	7.69		
Short Side	2	9.00	1.00	0.10	1.80		
				Total	112.99	249.26	28164.00
					Cum		

Rate Analysis

Basic Rate	242.00
Area waitage 3%	7.26
Net Rate Rs.	249.26 /Cum

SI No	Item of work	No	L	B	D	Qty	Rate Rs	Amount Rs
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- 3 Providing and laying in position cement concrete for levelling course for all works in foundation. The granite/trap/basalt crushed graded coarse aggregates and fine aggregates as per relevant IS Codes machine mixed, laid in layers not exceeding 150 mm thickness, well compacted using plate vibrators, including all lead & lifts, cost of all materials of quality, labour, Usage charges of machineries, curing, and all the other appurtenances required to complete the work as per technical specifications. **Mix 1:4:8(M5) Using 40 mm nominal size graded crushed coarse aggregates (KPWD S.R 2023-24, Page No.15, Item No.2.1.2)**

Wall Foundation

Long wall	2	36.00	1.00	0.15	10.80
Short wall	2	9.00	1.00	0.15	2.70

Below Flooring	1	36.45	9.45	0.10	34.44
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Flagging

Long Side	2	38.45	1.00	0.10	7.69
Short Side	2	9.00	1.00	0.10	1.80

Total	57.43	5833.92	335042.00
	Cum		

Rate Analysis

Basic Rate	5664
Area waitage 3%	169.92

Net Rate Rs. 5833.92 /Cum

- 4 Providing and laying in position cement concrete for all Foundation works. The granite/trap/basalt crushed graded coarse aggregates and fine aggregates as per relevant IS Codes machine mixed with super plasticizers laid in finished layers, well compacted using needle vibrators, including all lead & lifts, cost of all materials, quality confirming to the requirements of relevant IS codes, labour, Usage charges of machinery, curing and all the other appurtenances required to complete the work as per technical specifications. **M20 Design Mix Using 20 mm and down size graded crushed coarse aggregates. (KPWD-SR 2023-24; Vol 1; Page No 15; Item No. 2.2.1)**

Footing

Columns	26	0.80	0.80	1.00	16.64
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Wall Raft

Long wall-foot	2	36.00	0.70	0.30	15.12
Short wall-foot	2	9.00	0.70	0.30	3.78
Columns	26	0.80	0.30		-6.24

Total	29.30	6978.57	204472.00
	Cum		

Rate Analysis

Basic Rate	6287.00
Centering at 8%	502.96
Area waitage 3%	188.61

Net Rate Rs. 6978.57 /Cum

SI No	Item of work	No	L	B	D	Qty	Rate Rs	Amount Rs
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- 5 Providing and laying in position cement concrete for all Foundation works. The granite/trap/basalt crushed graded coarse aggregates and fine aggregates as per relevant IS Codes machine mixed with super plasticizers laid in finished layers, well compacted using needle vibrators, including all lead & lifts, cost of all materials, quality confirming to the requirements of relevant IS codes, labour, Usage charges of machinery, curing and all the other appurtenances required to complete the work as per technical specifications .M20 Design Mix Using 20 mm and down size graded crushed coarse aggregates. **(KPWD-SR 2023-24; Vol 1; Page No 15; Item No. 2.2.1)**

above gl

Columns	26	0.45	0.45	0.60	3.15
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Vertical wall

Long wall	2	36.00	0.30	1.30	28.08
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Short wall	2	9.00	0.30	1.30	7.02
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Deductions

Columns	26	0.45	0.30	1.30	-4.56
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Total					33.69	9619.11	324068.00
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Cum

Rate Analysis

Basic Rate 6287.00

Centering at 50% 3143.50

Area waitage 3% 188.61

Net Rate Rs. 9619.11 /Cum

- 6 Providing and laying in position cement concrete for all Super structures of building , Road works, Water works, Irrigation works & super structure works of bridges upto 3.50 m height. The granite/ trap/basalt crushed graded coarse aggregates and fine aggregates as per relevant IS Codes machine mixed with super plasticisers laid in layers, well compacted using needle vibrators. The cost includes all lead & lifts, cost of all materials, quality confirming to the requirements of relevant IS codes , labour, Usage charges of machinery, curing and all other appurtenances required to complete the work as per technical specifications. **M20 Design Mix Using 20 mm nominal size graded crushed coarse aggregates .(KPWD S.R 2023-24, Val - 1 Page No.16, Item No.2.5.1)**

Flooring	1	36.45	9.45	0.15	51.66
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Step	4	3.00	0.90	0.15	1.62
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	4	3.00	0.60	0.15	1.08
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	4	3.00	0.30	0.15	0.54
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Flagging

Long Side	2	38.45	1.00	0.10	7.69
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Short Side	2	9.00	1.00	0.10	1.80
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Total					64.39	7491.90	482403.00
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Cum

SI No	Item of work	No	L	B	D	Qty	Rate Rs	Amount Rs
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Rate Analysis

Basic Rate	6630.00
Centering at 10%	663.00
Area waitage 3%	198.9

Net Rate Rs. 7491.90 /Cum

- 7 Supplying, fitting and placing TMT FE 550 / 550D Steel Reinforcement including cost of all materials, machinery, labour, cleaning, straightening, cutting, bending, hooking, laping/welding joints, tying with binding wire / soft annealed steel wire and other ancillary operations complete as per drawing and technical specification **(KPWD-SR 2023-24; Vol 1; Page No 18; Item No. 2.11 a)**

For Columns Raft			at55 kg/Cum			915.20		
Foundation			at65 kg/Cum			1904.50		
Basement walls			at55 kg/Cum			1852.95		
					Kg/Sqm			
Flooring	1	36.45	9.45		3.50	1205.58		
						5878.23		
					Total	5.88	92140.71	541624.00
						T		

Rate Analysis

Basic Rate	89457.00
Area waitage 3%	2683.71

Net Rate Rs. 92140.71 /T

- 8 Providing 12 mm cement plaster with cement mortar 1:6 (1 cement: 6 fine sand) including rounding off corners wherever required smooth rendering, providing and removing scaffolding, including cost of materials, labour, curing complete as per specifications and as per directions of Engineer-in-charge. **(KPWD SR:2023-24, Vol-2 Page No. 41, Item No.8.1.2)**

Long wall	2	36.45		0.60	43.74			
Short wall	2	9.45		0.60	11.34			
				Total	55.08	248.23	13673.00	
					Sqm			

Rate Analysis

Basic Rate	241.00
Area waitage 3%	7.23

Net Rate Rs. 248.23 /Sqm

- 9 Construction of Embankment by excavating the available approved Gravel/Murum deposited at a place or borrow pits during or prior excavation with all lifts and lead, transportation to site, spreading, grading to required slope and compacting to meet the requirement complete as per specifications, including cost of labour,rolling,water,all materials,usage& all other appurtenances required to complete the work. **(KPWD SR:2023-24, Vol-1, Page No. 7. Item No.1.8)**

Inside Basement	1	35.55	8.55	0.30	91.18			
				Total	91.18	488.22	44516.00	
					Cum			

SI No	Item of work	No	L	B	D	Qty	Rate Rs	Amount Rs
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Rate Analysis

Basic Rate 474.00

Area waitage 3% 14.22

Net Rate Rs. 488.22 /Cum

- 10 Filling available excavated earth (excluding rock) in trenches, plinth, sides of foundations and other similar works etc. in layers not exceeding 20cm in depth, consolidating each deposited layer by ramming and watering, lead up to 50 m and lift upto 1.5 m. **(KPWD SR:2023-24, Vol-1, Page No. 7, Item No.1.9)**

Sundry Shops

Inside Foundation &

55.56		
<u>55.56</u>	273.98	15222.00
Cum		

Rate Analysis

Basic Rate 266.00

Area waitage 3% 7.98

Net Rate Rs. 273.98 /Cum

- 11 Providing and fixing Structural Steel work riveted, bolted or welded in built up sections, trusses and framed work, including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer all complete including cost of materials, labour, usage charges of machinery complete as per specifications and as per directions of the Engineer-in-Charge. **(KPWD SR:2023-24, Vol-2, Page No. 77, Item No.11.2)**

Columns-is-100x100x4	26	46.92	1219.92		
Truss-ISA 50x50x6	13	249.57	3244.41		
Purlins					
80x40*2.9 mm	12	185.61	2227.28		
Base Plate	104	4.24	440.85		
for gusset plates	18	4.00	72.00		
			<u>7204.46</u>	116.39	838527.00
			Kg		
			Total		

Rate Analysis

Basic Rate 113.00

Area waitage 3% 3.39

Net Rate Rs. 116.39 /kg

SI No	Item of work	No	L	B	D	Qty	Rate Rs	Amount Rs
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- 12 Providing and fixing precoated galvanised iron profile sheets (size, shape and pitch of corrugation as approved by Engineer-in-charge) 0.50 mm (+ 0.05 %) total coated thickness with zinc coating 120 grams per m2 as per IS: 277, in 240 mpa steel grade, 5-7 microns epoxy primer on both side of the sheet and polyester top coat 15-18 microns. Sheet should have protective guard film of 25 microns minimum to avoid scratches during transportation and should be supplied in single length upto 12 m or as desired by Engineer-in-charge. The sheet shall be fixed using self drilling/self tapping screws of size (5.5x 55 mm) with EPDM seal, complete upto any pitch in horizontal/ vertical or curved surfaces, excluding the cost of purlins, rafters and trusses and including cutting to size and shape wherever required.

(KPWD SR:2023-24, Vol-2, Page No. 26, Item No.7.38)

Roof Slopes	2	37.35	5.35			399.64		
					Total	<u>399.64</u>	585.04	233805.00
						Sqm		

Rate Analysis

Basic Rate	568.00
Area waitage 3%	<u>17.04</u>
Net Rate Rs.	585.04 /Sqm

- 13 Providing and fixing precoated galvanised steel sheet roofing accessories 0.50 mm (+0.05 %) total coated thickness, Zinc coating 120 grams per m2 as per IS: 277, in 240 mpa steel grade, 5-7 microns epoxy primer on both side of the sheet and polyester top coat 15-18 microns using self drilling/ self tapping screws complete : **Ridges plain (500 - 600mm) (KPWD SR:2023-24, Vol-2, Page No. 27, Item No.7.39.1)**

Ridge	1	37.35				37.35		
					Total	<u>37.35</u>	464.53	17350.00
						mtr		

Rate Analysis

Basic Rate	451.00
Area waitage 3%	<u>13.53</u>
Net Rate Rs.	464.53 /mtr

SI No	Item of work	No	L	B	D	Qty	Rate Rs	Amount Rs
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- 14 Painting with synthetic enamel paint of approved brand and manufacture of required colour to give an even shade :Two coats on new work over an under coat of suitable shade with ordinary paint of approved brand and manufacture after thoroughly brooming the surface to remove all dirt, dust, mortar drops and foreign matter including preparing the surface even and sand paper smooth, cost of materials, labour complete as per specifications and as per directions of Engineer-in-charge.

(KPWD SR:2023-24, Vol-2, Page No. 48, Item No.8.41)

Truss	13	9.00	1.50	0.50		87.75		
columns	26	4	0.40			41.60		
Purlins		12	37.35	0.26		116.53		
Total						<u>245.88</u>	172.01	42294.00
						Sqm		

Rate Analysis

Basic Rate	167.00
Area waitage 3%	<u>5.01</u>
Net Rate Rs.	172.01 /Sqm

- 15 Providing and fixing 15 cm wide, 45 cm overall semi-circular plain G.S. sheet gutter with iron brackets 40x3mm size, bolts, nuts and washers etc., including making necessary connections with rain water pipes complete. (0.63 mm thick with zinc coating not less than 275 g/m²) **(Common SR of Buildings 2023-24),Vol- II, Item No :7.6, Page No :21)**

Gutter

	2	37.4				74.70		
Total						<u>74.70</u>	664.35	49627.00
						Rmtr		

Rate Analysis

Basic Rate	645.00
Area waitage 3%	<u>19.35</u>
Net Rate Rs.	664.35 rmtr

SI No	Item of work	No	L	B	D	Qty	Rate Rs	Amount Rs
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- 16 Supplying, laying and jointing HDPE pipes of specified grade and conforming to IS 4984-2016 with latest amendments and conveying to work site including loading and unloading at both destinations and rolling and lowering into trenches, laying true to line and jointing of pipes and specials with electrofusion welding, giving hydraulic test as per relevant ISS with all lead and lifts including encasing the pipe alround to a depth of not less than 15 cms. with soft gravel or selected earth available from the excavation, testing and commissioning. The rate is exclusive of required specials and fittings wherever necessary like saddle Tee, stub ends, flanged sets, bends, reducers etc. complete (Contractor will make his own arrangements for procuring water for testing) etc. Note: Upto 110mm dia Coil shall be used. **For Grade PE80 PN6.0,(Common SR of Water Supply & UGD 2023-24),Vol-V, Item No :9.1.4, Page No :95)**

Rain Water pipe

1	4.0	13.00			52.00			
			Total		<u>52.00</u>		483.07	25120.00
					Rmtr			

Rate Analysis

Basic Rate	469.00	
Area waitage 3%	<u>14.07</u>	
Net Rate Rs.	483.07	rmtr

Total Rs 3207155.00

Section Engineer

Assistant Executive Engineer
No. 6 Sub-Division, MDP,
Arsikere.

Executive Engineer
MDP,Mysore Division
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JNANA BHARATHI
BANGALORE-560 056

BUB/2023-24/SBC-

Date: 14-09-2023

TEST REPORT ON SBC OF SOIL

Sub: Determination of SBC of soil sample
{Project*: Construction of CC.Road, RCC Drain & CAP at Didiga SMY of APMC
Channarayapatna}

Ref: Testing/BUB/2023-24, dated 09-09-2023

Samples supplied to Lab by: M/s. Geopage Consultants,
#104/22, 6th Cross, KHB Colony,
Basaveshwaranagar, Bengaluru-560079.

Technical reference: As per IS-2720 p-13 & IS 6403-1981.

TABLE 1 SHEAR TEST RESULTS

Sl. No	Sample Identification *	Depth of sampling* m	Density t/m ³	Shear Parameters	
				Cohesion t/m ²	Angle of Internal Friction, Degrees
1	Sample 01- Covered Auction Platform	1.50	1.890	1.18	28.0°

* As furnished by the Client

* The Results relate only to the items tested only

Safe Bearing Capacity:

Based on the test results obtained by conducting laboratory tests on the soil sample, supplied by the client, a Safe Bearing capacity of **13.80t/m²** is recommended at a depth of 1.50 m below ground level, with a factor of safety of 3.0 against shear failure.



Page No 1/2

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CALCULATION OF SAFE BEARING CAPACITY

{Project*: Construction of CC.Road, RCC Drain & CAP at Didiga SMY of APMC Channarayapatna}

TYPICAL CALCULATION OF ALLOWABLE BEARING PRESSURE

Soil Parameters :

At a depth of 1.50m, $c = 1.18 \text{ t/m}^2$, $\phi = 28.0^\circ$, and $\gamma = 1.890 \text{ t/m}^3$
Considering local shear failure, $\phi_m = \tan^{-1}(0.67 \tan \phi) = 19.60$

SHEAR FAILURE CONDITION

Bearing capacity factors:

$$N_q = e^{\pi \tan \phi} \tan^2 (45 + \phi_m/2) \\ = 6.06$$

$$N_c = (N_q - 1) \cot \phi_m \\ = 14.36$$

$$N_\gamma = 2 (N_q + 1) \tan \phi_m \\ = 4.95$$

Depth factors:

$$d_c = 1 + 0.2 (D_f/B) \tan (45 + \phi/2) \\ = 1.499$$

$$d_q = d_\gamma = 1 + 0.1 ((D_f/B) \tan (45 + \phi/2)) \\ = 1.250$$

Shape factors: $S_c = 1.2$, $S_q = 1.3$, $S_\gamma = 0.8$

Substituting the above factors in,

$$Q_r = \frac{2}{3} c N_c S_c d_c + \gamma' D (N_q - 1) S_q d_q + 0.5 \gamma B N_\gamma S_\gamma d_\gamma$$

$$Q_r = 48.30 \text{ t/m}^2$$

Using a factor of safety of 3.0, $Q_a = 13.80 \text{ t/m}^2$



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S.A.K. 24.8.2023
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BUB/2023-24/SBC-

Date: 14-09-2023

TEST REPORT ON SBC OF SOIL

Sub: Determination of SBC of soil sample

{Project*: Construction of CC.Road, RCC Drain & CAP at Didiga SMY of APMC
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Ref: Testing/BUB/2023-24, dated 09-09-2023

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Technical reference: As per IS-2720 p-13 & IS 6403-1981.

TABLE 1 SHEAR TEST RESULTS

Sl. No	Sample Identification *	Depth of sampling* m	Density t/m ³	Shear Parameters	
				Cohesion t/m ²	Angle of Internal Friction, Degrees
1	Sample 02- Covered Auction Platform	1.50	1.882	1.10	28.0°

* As furnished by the Client

* The Results relate only to the items tested only

Safe Bearing Capacity:

Based on the test results obtained by conducting laboratory tests on the soil sample, supplied by the client, a Safe Bearing capacity of **13.68t/m²** is recommended at a depth of 1.50 m below ground level, with a factor of safety of 3.0 against shear failure.



Page No 1/2

S. A. Khan 14.09.2023
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CALCULATION OF SAFE BEARING CAPACITY

{Project*: Construction of CC.Road, RCC Drain & CAP at Didiga SMY of APMC Channarayapatna }

TYPICAL CALCULATION OF ALLOWABLE BEARING PRESSURE

Soil Parameters :

At a depth of 1.50m, $c = 1.10 \text{ t/m}^2$, $\phi = 28.0^\circ$, and $\gamma = 1.882 \text{ t/m}^3$
Considering local shear failure, $\phi_m = \tan^{-1}(0.67 \tan \phi) = 19.60$

SHEAR FAILURE CONDITION

Bearing capacity factors:

$$N_q = e^{\pi \tan \phi} \tan^2 (45 + \phi_m/2) \\ = 6.20$$

$$N_c = (N_q - 1) \cot \phi_m \\ = 14.52$$

$$N_\gamma = 2 (N_q + 1) \tan \phi_m \\ = 5.17$$

Depth factors:

$$d_c = 1 + 0.2 (D_f / B) \tan (45 + \phi/2) \\ = 1.499$$

$$d_q = d_\gamma = 1 + 0.1 ((D_f / B) \tan (45 + \phi/2)) \\ = 1.250$$

Shape factors: $S_c = 1.2, S_q = 1.3, S_\gamma = 0.8$

Substituting the above factors in,

$$Q_f = \frac{2}{3} c N_c S_c d_c + \gamma' D (N_q - 1) S_q d_q + 0.5 \gamma B N_\gamma S_\gamma d_\gamma$$

$$Q_f = 47.89 \text{ t/m}^2$$

Using a factor of safety of 3.0, $Q_a = 13.68 \text{ t/m}^2$

